Complete measurement inputs for the mix criteria

Human Compression project — F005 input audit

This packet fixes the text and program boundaries for one complete historical article: Cunningham and Hubard, *Polytopality criteria for the mix of polytopes and maniplexes*, arXiv:2506.19334v1, 24 June 2025. All 21 physical pages are included. The comparison will use the entire common paper, the entire local application and the entire exceptional-case addition. No part of that addition is free prior work, and there is no overlap or cross-topic deduction.

The primary coordinate was fixed before counting: NFC characters without whitespace in the literal, untrimmed PDF extraction. All original and replacement documents use this same rule. Page footers and the source's arXiv banner therefore remain in the primary counts. No ratio is evaluated here.

Whole document	Pages	Raw	Selected	Calibrated
Complete historical article	21	57,753	57,720	57,720
Common theory	5	12,615	12,610	12,610
Local application	8	20,788	20,780	20,780
Exceptional two-orbit application	4	10,289	10,285	10,285
Main complete program appendix	4	8,813	8,809	8,809
Exceptional complete program appendix	3	7,212	7,209	7,209

The last three columns show NFC characters without whitespace. "Selected" removes only each isolated physical page footer. "Calibrated" additionally substitutes individually inspected visible glyphs, leaving the same number of nonspace characters here but sometimes changing their UTF-8 cost. Neither column changes the primary boundary. The complete machine-readable inventory records five units, NFC and NFKC, and all three variants: 180 PDF cells across six documents. The five units are whitespace tokens, alphanumeric units, characters without whitespace, characters with collapsed whitespace, and UTF-8 bytes with collapsed whitespace.

Scientific scope. The bound main packet has 35 detailed source rows; these are not 35 originals. Its additive application settles the literal claims recorded as R23 and R29 by counterexamples and gives precise rooted replacement criteria. All three final source problems remain unanswered and excluded from the recovery claim. Other corrections, qualifications and retained inputs remain exactly those of the bound scientific records. The measurement author also authored the mathematics; distinct scientific acceptance is now bound to both unchanged scientific editions. Distinct measurement-input review is still required before factors may be calculated.

Whole program alternatives. The two complete printed appendices have seven pages in total. Their raw alternative consists of the three whole mathematical files listed on the next page. Printed and raw forms are alternatives: they must never be added together. Any later conditional application comparison must declare the whole common theory and all three mathematical programs already paid. Drivers and partial formal evidence are inventoried separately, without silently folding them into those programs.

Executable boundaries and representation limits

Whole raw file	Role	NFC nonspace
`check_coprime.g`	Required finite certificate	1,871
`check_single_base.py`	Six-coordinate validation	6,127
`check_exceptional.py`	Exceptional affine validation	6,610
`check.py`	Main replay driver	7,238
`check_integrity.py`	Integrity controls	1,827
`check.py`	Additive replay driver	4,609
`FiberIntersections.lean`	Scoped formal file	1,714

The GAP file is a required finite certificate for the eleven-cell and simplex markings in the coprime-label obstruction. Both Python mathematical files validate complete written coordinate arguments; they do not replace the uniform proofs. Full-file inclusion in the two existing appendices is checked against the literal source bytes, with no line pruning or abbreviated output. All seven raw files have both normalizations and all five units recorded (70 cells). The Lean file contains two generic fiber declarations; this is not an end-to-end formalization of the geometry, finite groups or source.

Actual fonts and pages. The author viewed all 45 scientific and code pages and 27 exceptional embedded glyph outlines. The source has 23 actual Type1 font dictionaries and 469 encoded glyph classes. One replacement code appendix also has a Type1 dictionary with five classes; 18 replacement Type0 dictionaries retain their literal extraction. Nine visually identified source aliases occur 1,210 times. They affect only the calibrated sensitivity. In particular a decoded mapsto component is retained as an inspected bar-and-stem glyph, not reconstructed into a guessed composite mathematical arrow.

The extractor pins `pypdf 6.13.0` and `fontTools 4.63.0`. It checks the returned raw text against ordinary uninstrumented extraction, records the original font and byte events for every substitution, and keeps generated line breaks separate from actual glyph events. There is no Form callback duplication in these inputs. Forty rejection controls exercise the whole-document boundary, code roles and completeness, forbidden discounts, source qualifications, literal extraction and glyph substitutions.

What the counts do not measure. Source diagrams on pages 9 and 11 and displayed path relations on page 14 contain graphical information. Their extractable labels and literal text proxies remain counted; strokes, color and spatial placement have no established full cost. The historical software, retained external proofs and full formalization costs are likewise unknown, not zero. The figures have not been replaced by invented textual equivalents. The font substitutions are a representation sensitivity, not an estimate of these missing costs.

The portable replay accepts explicit source and public-artifact roots and verifies complete PDF, program and scientific-manifest hashes. It exports no historical article text, font bytes or page images. Optional isolated TeX rebuilds check the reader and both complete code appendices; the input audit does not rerun the mathematical GAP or Lean proofs. Whole-source scientific coverage, measurement-input acceptance and a subsequent numerical comparison remain separate decisions.